

Lucy Coe

✉ lucycoe07@gmail.com | 🌐 GitHub | 🔗 LinkedIn | 🌐 Portfolio | 📄 StudyFlow

Education

BSc Computer Science, University of Edinburgh

2024–2028

Predicted First Class Honours

Expected Graduation: July 2028

Relevant Modules: Algorithms & Data Structures, Computer Systems, Software Engineering and Professional Practice, Reasoning and Agents, Computer Security

Experience

Object-Oriented Programming Tutor and Marker

2026

University of Edinburgh

- Led weekly Java/OOP tutorials for 15+ students, communicating technical concepts clearly, providing coursework feedback and explaining debugging, object-oriented design and code quality.

Cognitive Science Python Lab Demonstrator

2026

University of Edinburgh

- Guided 5–15 students through weekly Python labs, troubleshooting code and explaining NumPy and Jupyter-based computational exercises.

Customer Service and Hospitality Assistant

Until 2024

Café, Restaurant and Pottery Studio

- Worked across café, restaurant and pottery studio roles in customer-facing environments, independently managing pottery studio shifts alongside customer service and sales responsibilities.

Projects

Simian Engine

Python, FastAPI, React, TypeScript, PixiJS, Docker, PyTorch

- Built an agent-based simulation exploring emergent behaviour among autonomous agents across a procedurally generated 300×300 environment, with an interactive PixiJS visualisation.
- Engineered autonomous behaviour for 300+ agents using A* pathfinding and spatial memory, integrating a PyTorch neural-network controller with Q-values and reward-based learning.
- Profiled simulation performance across neural-network training, pathfinding and per-tick updates, identifying bottlenecks and refactoring critical components as the system scaled.
- Developed a FastAPI backend for the React frontend and containerised both services with Docker Compose.

HTTP/1.1 Server

Java, TCP Sockets, Maven, JUnit 5

- Built an HTTP/1.1 server from raw TCP sockets without a web framework, implementing request parsing, dynamic routing, middleware, static file serving, cookies and server-side sessions.
- Designed a concurrent worker pool, graceful shutdown and path-traversal protection to improve reliability under concurrent requests; validated HTTP/TCP behaviour with 119 unit and integration tests.

StudyFlow

React, Supabase, PostgreSQL, Vitest

- Built and deployed a full-stack study management platform using React, Supabase and PostgreSQL, integrating coursework, timetables, flashcards and task management into a single application.
- Designed a constraint-aware scheduling algorithm to prioritise coursework by deadline and workload while avoiding timetable conflicts; secured user data with row-level security and validated functionality with 230 automated tests.

Cache Simulator

C, Assembly

- Implemented a configurable cache simulator in C supporting direct-mapped, set-associative and fully associative caches, LRU replacement and two-level L1/L2 hierarchies with hit/miss instrumentation.

Student Event Booker

Java, JUnit

- Collaborated in a four-person Agile team from requirements and design through implementation and testing, using Git and peer review to deliver a Java event-booking system validated by 100+ JUnit tests.

Technical Skills

Languages: Python, Java, SQL, TypeScript, JavaScript, C

Backend: FastAPI, REST APIs, PostgreSQL, Supabase, HTTP, TCP/IP

Frontend: React, PixiJS, HTML, CSS, Tailwind CSS

Testing: JUnit 5, Vitest, React Testing Library

Tools: PyTorch, Git, Docker, Maven, Linux (WSL), Vercel, AWS

Awards and Achievements

Amazon Future Engineer Scholarship • Robertson Trust Scholarship • Proxime Accessit (Wick High School)